PROJ IS QUASI-COMPACT EXACTLY WHEN THE IRRELEVANT IDEAL HAS A FINITELY GENERATED RADICAL

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Topology of Proj. algebra-lemma-topology-proj (cross-chapter, named). Standard opens form a basis, and D_+(f_i) cover Proj(S) exactly when S_+ sits in the radical of the ideal generated by the f_i.

Source reference. Original reference 00JP: focused-seeds/geometry/S9-cons-proj-quasi-compact/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a graded ring. The scheme Proj(S) is quasi-compact if and only if there exist finitely many homogeneous elements $f_1, \ldots, f_n \in S_+$ such that $S_+ \subset \sqrt{(f_1, \ldots, f_n)}$. In this case Proj$(S) = D_+(f_1) \cup \ldots \cup D_+(f_n)$.*

Proof. Given such a collection of elements the standard affine opens $D_+(f_i)$ cover $\text{Proj}(S)$ by Algebra, Lemma 00JP. Conversely, if $\text{Proj}(S)$ is quasi-compact, then we may cover it by finitely many standard opens $D_+(f_i)$, $i = 1, \ldots, n$ and we see that $S_+ \subset \sqrt{(f_1, \ldots, f_n)}$ by the lemma referenced above. □

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.